Che Chang

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Technical Skills

Programming: C++, CUDA | **Specialties:** GPU Programming, Parallel Algorithms, High-Performance Computing, Graph Algorithms, Static Timing Analysis

Education

University of Wisconsin–Madison Madison, WI
Ph.D. in Electrical and Computer Engineering; GPA: 3.5 *Sep. 2023 – Dec. 2026 (Expected)*

University of Utah Salt Lake City, UT
Master of Entertainment Arts and Engineering; GPA: 3.8 *Aug. 2020 – May 2022*

National Cheng Kung University Tainan, Taiwan
B.S. in Computer Science *Sep. 2015 – Jun. 2019*

Experience

Cadence Design Systems San Jose, CA
Software Engineer Intern, System Verification Group *May 2026 – Aug. 2026*

- Designed algorithms to optimize cross-processor connection latency, accelerating hardware emulation by **10–20%**.

Google Mountain View, CA
Student Researcher, AI^2 Chips Team *May 2024 – Aug. 2024*

- Developed parallel algorithms for SPICE simulation and pruned redundant input patterns, achieving up to **20× speedup** in timing analysis across multiple designs.

Intel Corporation Santa Clara, CA
Software Engineer Intern, Design Methodology and Automation Team *May 2022 – Aug. 2022*

- Automated FPGA design debugging with a constraint analyzer, improving the efficiency of design-rule diagnosis.

Selected Publications

G-PathGen: An Efficient GPU-Parallel Critical Path Generation Algorithm *ICS 2026*

- Designed CUDA GPU kernels for parallel critical-path generation and adaptive path-count control to maximize GPU utilization while minimizing redundant work.
- Achieved **1.6–243.8× speedup** over a state-of-the-art GPU method with **100% accuracy** when generating one million paths on large designs.

PathGen: An Efficient Parallel Critical Path Generation Algorithm *ASP-DAC 2025*

- Built a CPU-parallel algorithm using slack-based path grouping and dynamic load rebalancing to reduce thread contention and improve memory efficiency.
- Achieved **2.7–7.4× speedup** with 16 threads and nearly **100% accuracy**; selected as a **Best Paper Nominee** (13 of 169 accepted papers).

Ink: Efficient Incremental k-Critical Path Generation *DAC 2024*

- Designed an incremental algorithm that reuses paths across timing queries and prunes the critical-path search space.
- Achieved **5.2–22.4× speedup** and **20–31% lower memory usage** than a state-of-the-art timer on one million paths.

G-PASTA: GPU-Accelerated Partitioning Algorithm for Static Timing Analysis *DAC 2024*

- Developed a CUDA-accelerated algorithm that partitions large timing-dependency graphs into smaller subgraphs to reduce processor scheduling overhead.
- Delivered up to **41.8× faster partitioning** and reduced total static timing analysis runtime by **43%** on large designs.